

Disease Transmission in Dental Practice & Vaccinations

OMFS 513: Integrated Biomedical and Clinical Sciences 2

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Mission: To advance the science and art of health through education, service, scholarship and social accountability

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Learning Outcomes & Competency Statements

Learning Outcomes

1. Review the different modes of disease transmission.
2. Compare bloodborne pathogens to respiratory pathogens.
3. Review ways to prevent the transmissions of pathogens in dental practice.
4. Review the importance of vaccine administration.
5. Discuss new laws which allow dentist to administer vaccinations.
6. Discuss the pros and cons of vaccines in dental practice.

Competency statements met

1.3, 2.1, 3.1, 4.4, and 5.1

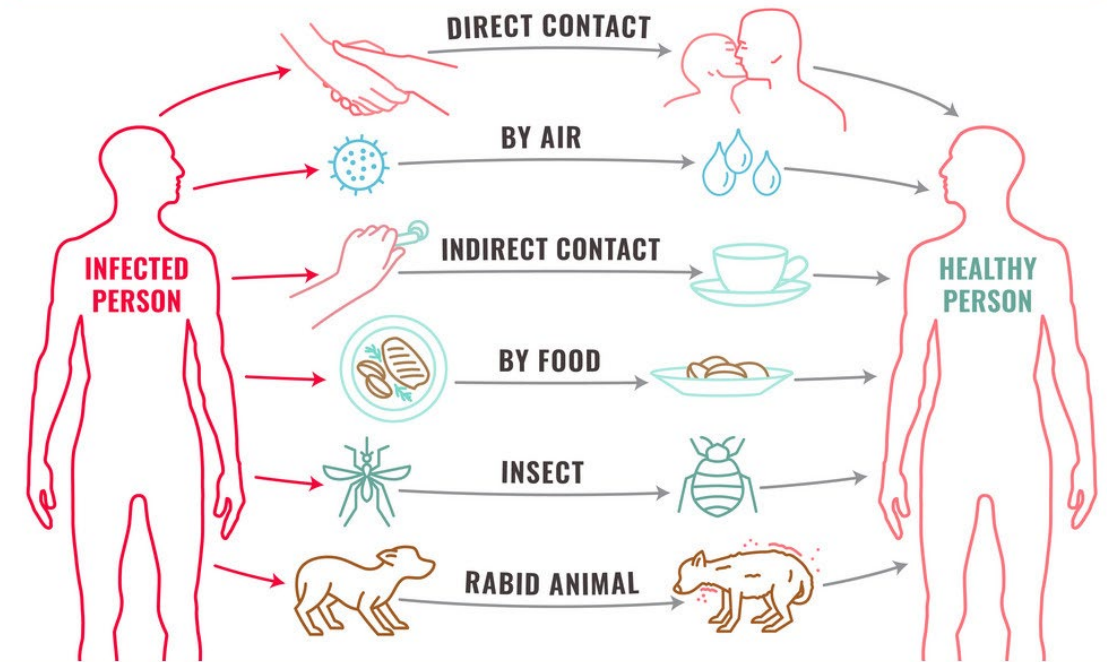
Governing Bodies

- **Occupational Safety and Health Administration (OSHA)**
 - Falls under the U.S Department of Labor
 - Enforces regulations to protect employers, employees, and patients from bloodborne pathogens.
 - Employers must provide annual OSHA training for all employees.
- **Center for Disease Control and Prevention (CDC)**
 - Falls under the Department of Health and Human Services
 - Creates infection control guidelines and recommendations for dentistry in addition to the promotion of global and public health.
- **Environmental Protection Agency (EPA)**
 - Regulates infectious waste disposal, environmental surveillance

Disease Transmission

- The spread of a communicable pathogens from an infected host to another individual or group.
- Direct
 - Transmission occurs through interpersonal contact
- Indirect
 - Transmission occurs through a common vehicle

TYPES OF DISEASE TRANSMISSION

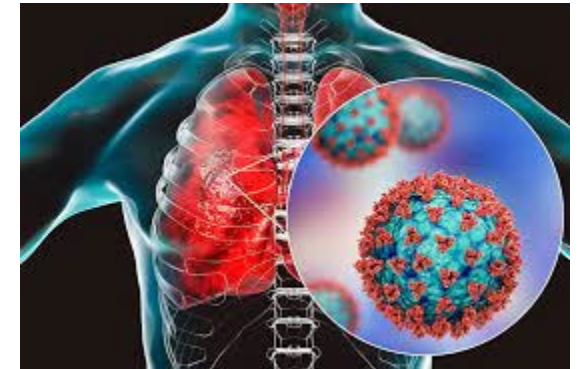
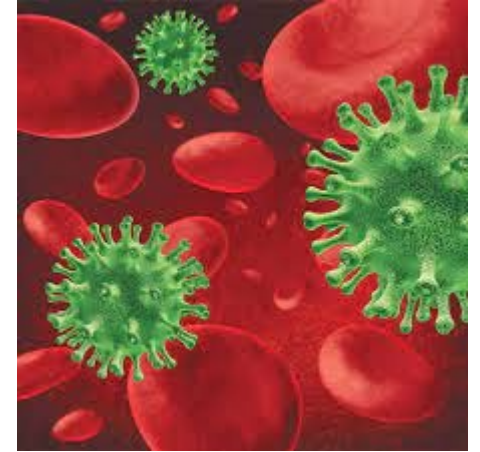


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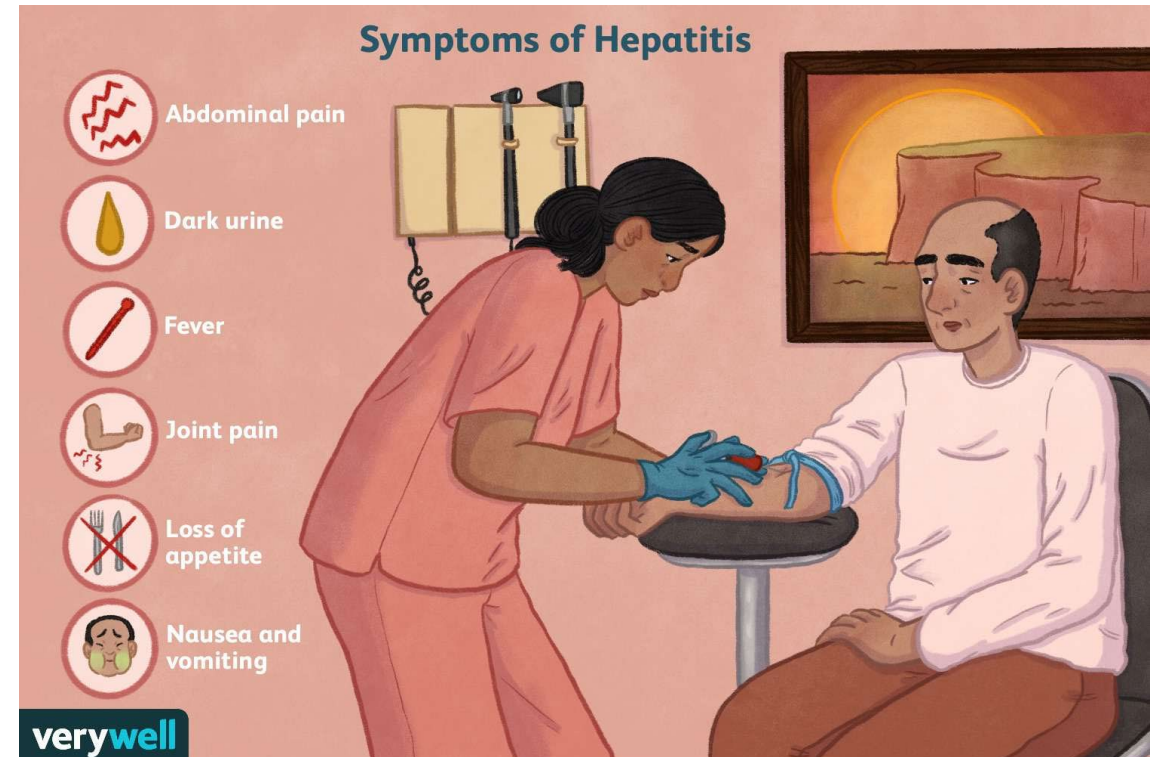
Diseases

- **Bloodborne**
 - Hepatitis B and C
 - HIV/AIDS
- **Respiratory**
 - Common cold
 - Influenza
 - Tuberculosis
 - SARS-CoV-2 (COVID-19)
- **Other Potentially Infectious Materials**

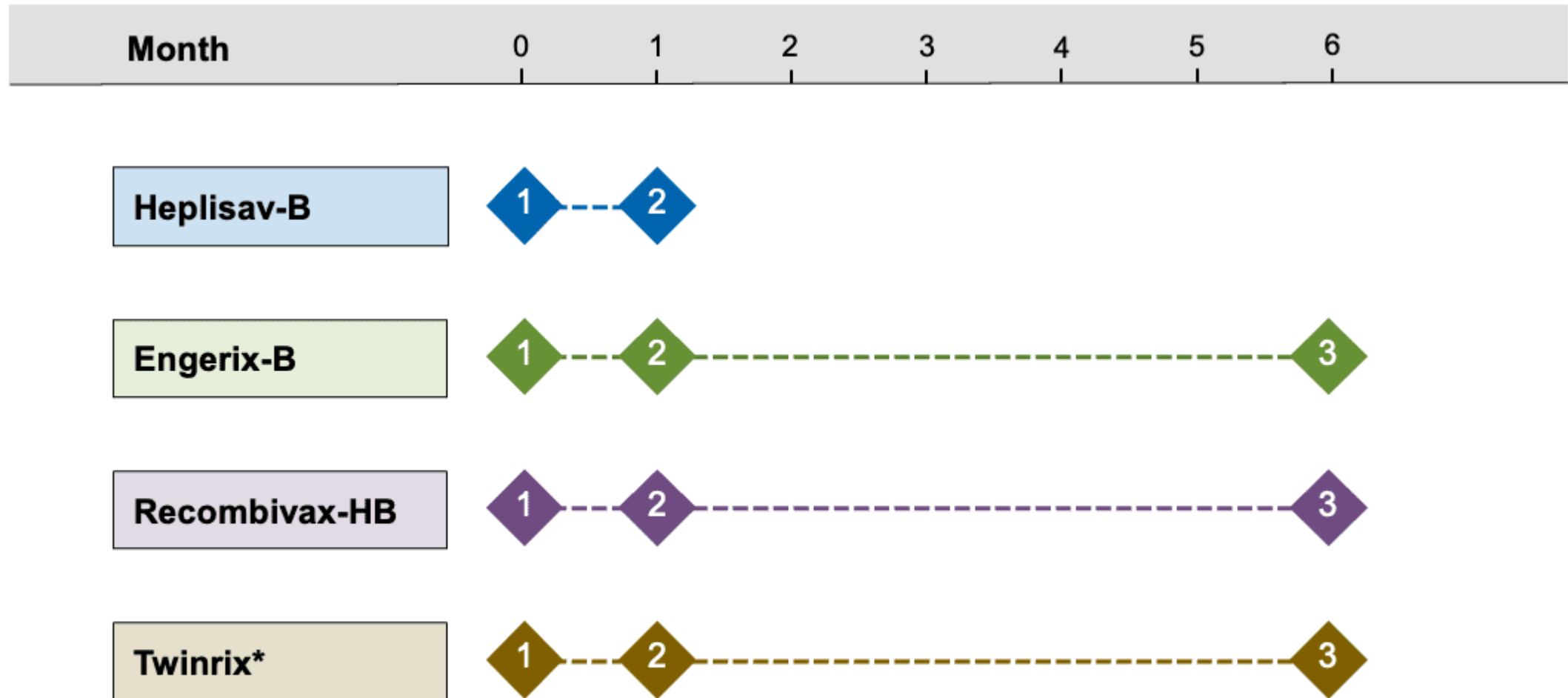


Hepatitis B

- Over 300 million currently living hepatitis B
- Acute disease
- Chronic disease
- Vaccine
 - 2 dose
 - 3 dose
 - 4 dose



Hepatitis B Vaccine Schedule - Adults



Hepatitis C

- Infectious viral disease of the liver
- Over 2.4 million Americans are living with the disease
- Acute vs chronic disease
- Age group most affected
 - Middle-aged and elderly adult
- Curable

Hepatitis C
What you should know

Tacoma-Pierce County
Health Department
Healthy People in Healthy Communities

What is it?
Hepatitis C is a virus that attacks the liver. Untreated, it can be life-threatening.

Can it be cured?
Yes! New treatments can cure hepatitis C for almost everyone.

It doesn't spread through...

How does it spread?

Used or shared needles, syringes or lancets

Sex

Sharing care items like toothbrushes or razors

Born to a mom with hepatitis C

Medical mishaps like needle injuries or unsterile instruments

Breastfeeding

Sharing eating utensils

Kissing, hugging or holding hands

Insect bites

Coughing or sneezing

What are the symptoms?

No symptoms

Stomach pain

Vomiting

Yellow eyes or skin

Am I at risk?

Go to www.tpchd.org/hepc and take a free online test to determine your risk.

People born between 1945 and 1965 are five times more likely to have hepatitis C.

To learn more about hepatitis C, go to www.tpchd.org/hepc.

Transmission: Bloodborne Pathogens

- **Direct exposure**
 - Sexual Contact
 - Mother-to-child exposure
- **Indirect exposure**
 - Blood transfusion
 - Intravenous drug use
 - Exposure to sharp instruments
 - Improperly sterilized instruments
 - Contaminate surfaces, waterlines, and medical appliances

Disease	Risk of acquiring disease after exposure
HIV	0.3%
Hepatitis C	3%
Hepatitis B	30%

Aerosol Transmissible Diseases

- Common cold, influenza, tuberculosis, and COVID-19
- Direct exposure
 - Aerosols
 - Bodily fluids
 - Mother to child
- Indirect exposure
 - Contaminated surfaces, waterlines, and medical appliances
 - Improperly sterilized instruments

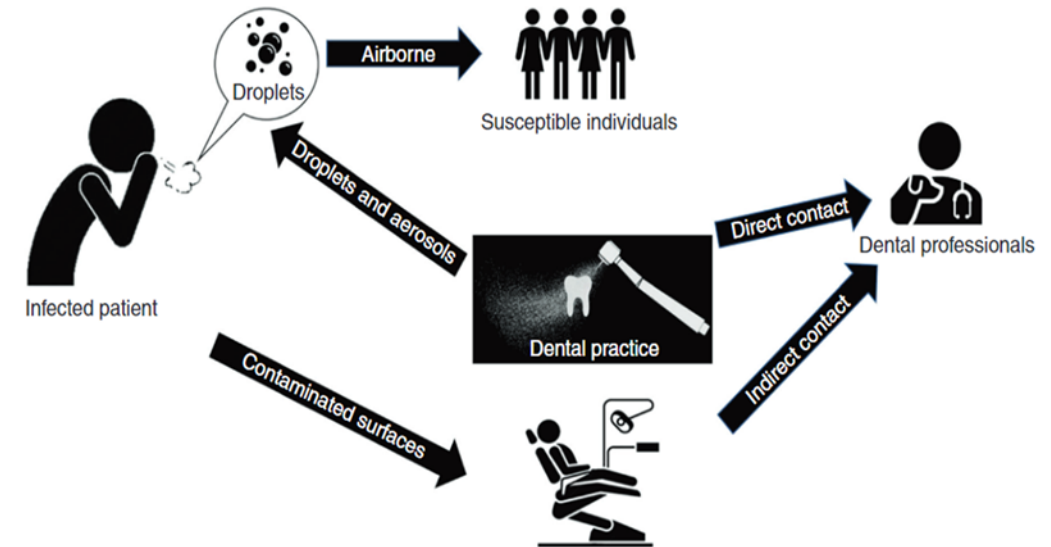


Fig. 1 Illustration of transmission routes of 2019-nCoV in dental clinics and hospitals

COVID-19

- SARS-CoV-2
- Highly transmissible respiratory virus
 - Symptoms
 - Fever, nasal congestion, pharyngitis, dry cough, dyspnea, and myalgias
 - Some patients experience GI upset
 - Skin manifestation in young children

COVID-19

- March 2020
 - COVID-19 is declared a national pandemic
 - States in the US begin to undergo “lock down”
 - Non-essential workers are asked to work from home
 - Dental clinics
 - Emergency treatment only
 - Teledentistry

Reopening Dental Offices in 2020

- COVID-19 Screening questionnaire
- Social distance in office
- New PPE requirements
- Pre-procedural mouth rinse
- Minimizing aerosol production
- Delaying treatment of patients with active disease
- Vaccination



Screening

- Temperature
- Screening Questionnaire
- Patient observation
- Reviewing medical history

	Questions	Yes	No
1	Do you feel feverish, have flu-like chills or have a fever with a temperature taken by mouth equal to or greater than 38.1°C (100.6°F)?		
2	Do you have a recent cough or a chronic cough that has recently gotten worse?		
3	Do you have difficulty breathing or are you short of breath?		
4	Have you had a sudden loss of sense of smell, without nasal congestion, with or without a loss of taste?		
5	Do you have a sore throat?		
6	Were you instructed to go into preventive isolation (e.g., after returning from travel abroad in the past 14 days ¹ , or having had contact with a person with confirmed COVID-19)?		
7	Was it recommended that you be tested for COVID-19 or are you waiting for a test result?		
8	Are you currently in isolation after testing positive for COVID-19?		
9	Were you in close contact (at least 15 minutes, at a distance of less than 2 metres) with a person with confirmed or suspected COVID-19, without wearing appropriate personal protective equipment, in the past 14 days?		
10	Do you have any of the following general symptoms?		
	• Diarrhea		
	• Nausea or vomiting		
	• Stomach ache		
	• Unusual severe fatigue for no obvious reason		
	• Severe loss of appetite		
	• Unusual generalized muscle pain or soreness (not related to physical effort)		
	• Unusual headaches		

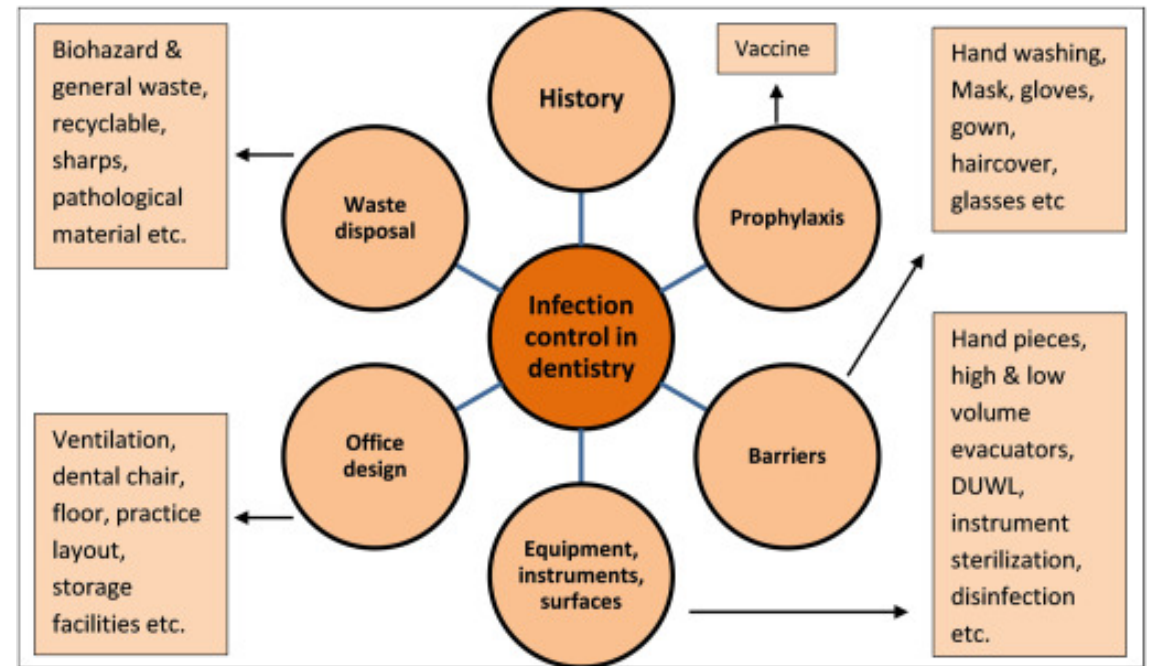
Anyone who answered “yes” to question 1, 2, 3, 4, 5, 6, 7, 8 or 9, or who answered “yes” to at least two symptoms in question 10 cannot be seen at the clinic. The person in question can be offered a teleconsultation or be referred to the most appropriate healthcare professional, based on his or her condition.

Other Potentially Infectious Material (OPIM)

- Human bodily fluids including **saliva that is visibly contaminated with blood**, or all body fluids where it is difficult or impossible to differentiate between fluids.
 - Examples: Semen, vaginal fluids, blood and other internal body fluids
- Unfixed tissue or organ, other than intact skin, from a human (living or dead).
- Cell cultures, tissue-cultures, solutions, organs or tissues from experimental animals contain HIV or HBV.

How to prevent transmission

- Medical history and screening
- Follow Standard Precautions
 - Personal protective equipment
 - Keep up to date with COVID-19 protocols
 - Handling sharps and soiled waste
 - Sterilization and disinfection
 - Proper use of barriers
 - Hand washing
 - Proper ventilation systems



Handwashing

5.5.4.1 Hand Hygiene Technique with Alcohol-Based Formulation

(Appendix 1 & 2)

- A. Duration of the entire procedure: 20 seconds at least.
- B. Apply a palm full of alcohol-based hand rub and cover all surfaces of the Hands
- C. Rub hands palm to palm
- D. Right palm over left dorsum with interlaced fingers and vice versa
- E. Palm to palm with fingers interlaced
- F. Backs of fingers to opposing palms with fingers interlocked
- G. Rotational rubbing of left thumb clasped in right palm and vice versa

5.5.4.2 Hand Hygiene Technique with Soap and Water

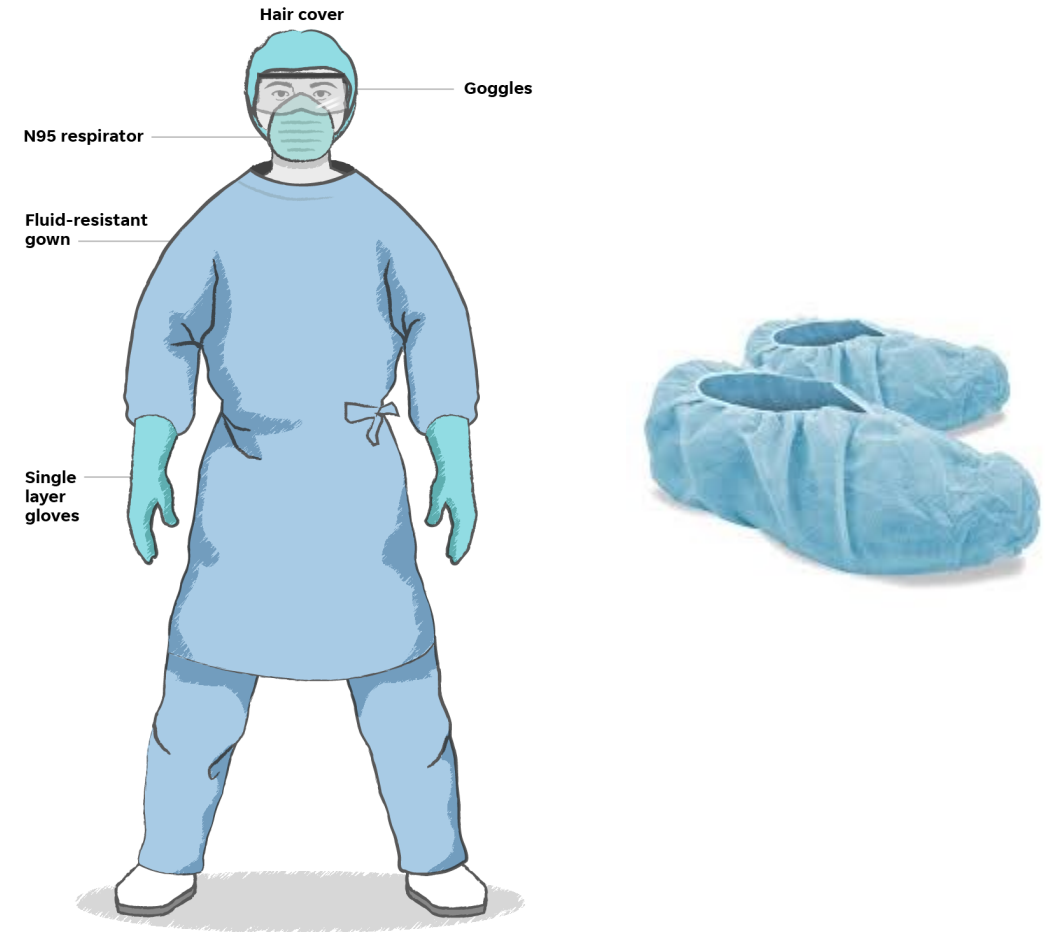
(Appendix 1 & 2)

- A. Duration of the entire procedure: 40-60 seconds
- B. Wet hands with water
- C. Apply enough soap to cover all hand surfaces
- D. Rub hands palm to palm
- E. Right palm over left dorsum with interlaced fingers and vice versa
- F. Palm to palm with fingers interlaced
- G. Backs of fingers to opposing palms with fingers interlocked
- H. Rotational rubbing of left thumb clasped in right palm and vice versa
- I. Rinse hands with water.
- J. Dry hands thoroughly with a single-use paper towel
- K. Use towel to turn off faucet



Personal Protective Equipment

- Eye wear
- Gloves
- Protective outer covering
 - Hair covering
 - Feet coverings
 - Surgical gowns
- Face shield
- Surgical mask
- N-95 mask



Sterilization and disinfection

- Written guidelines for sterilization and disinfection
- Assign responsibilities to all staff members
- Remove blood and other debris from the instruments prior to placement in the autoclave
- Use the proper disinfection mixtures to clean all operatories and other contaminated surfaces before and after treatment
- Flush the waterlines in all operatories
- Cover exposed surfaces in the operator to prevent contamination

HOW TO PROPERLY SEGREGATE DENTAL WASTE

MEDICAL	Blood-saturated and visibly contaminated items can pose a significant risk of transmitting infection. These items should be placed in red biohazard bags and treated as regulated medical waste. Fill It: Use Medical Waste Management mailback containers to dispose of your regulated (red bag) medical waste.  • Blood-Saturated Items • Visibly Contaminated PPE • Blood and Body Fluids • Blood-Saturated Gauze • Blood-Saturated Bandages
SHARPS	A good rule of thumb is if the item can puncture the skin and possibly contain blood borne pathogens, then it should be disposed of within a sharps container. Fill It: Use a Sharps Management mailback container to dispose of your sharps waste.  • Blades and Lancets • Needles • Carpules with visible blood or broken • Extracted teeth (no amalgam fillings) • Used syringes
PHARMACEUTICAL	Any unused or partially used medications, whether in tablet, capsule, or liquid form should be considered pharmaceutical waste. This also includes the items that contain the medications. Fill It: Use a Pharmaceutical Recovery mailback container to dispose of your pharmaceutical waste.  • Unused and partially used medications • Unused auto-injectors • Pharmaceutical vials and inhalers • Carpules (no blood)

Solid Medical Waste

- Blood-soaked gauze, contaminated PPE are

Sharps

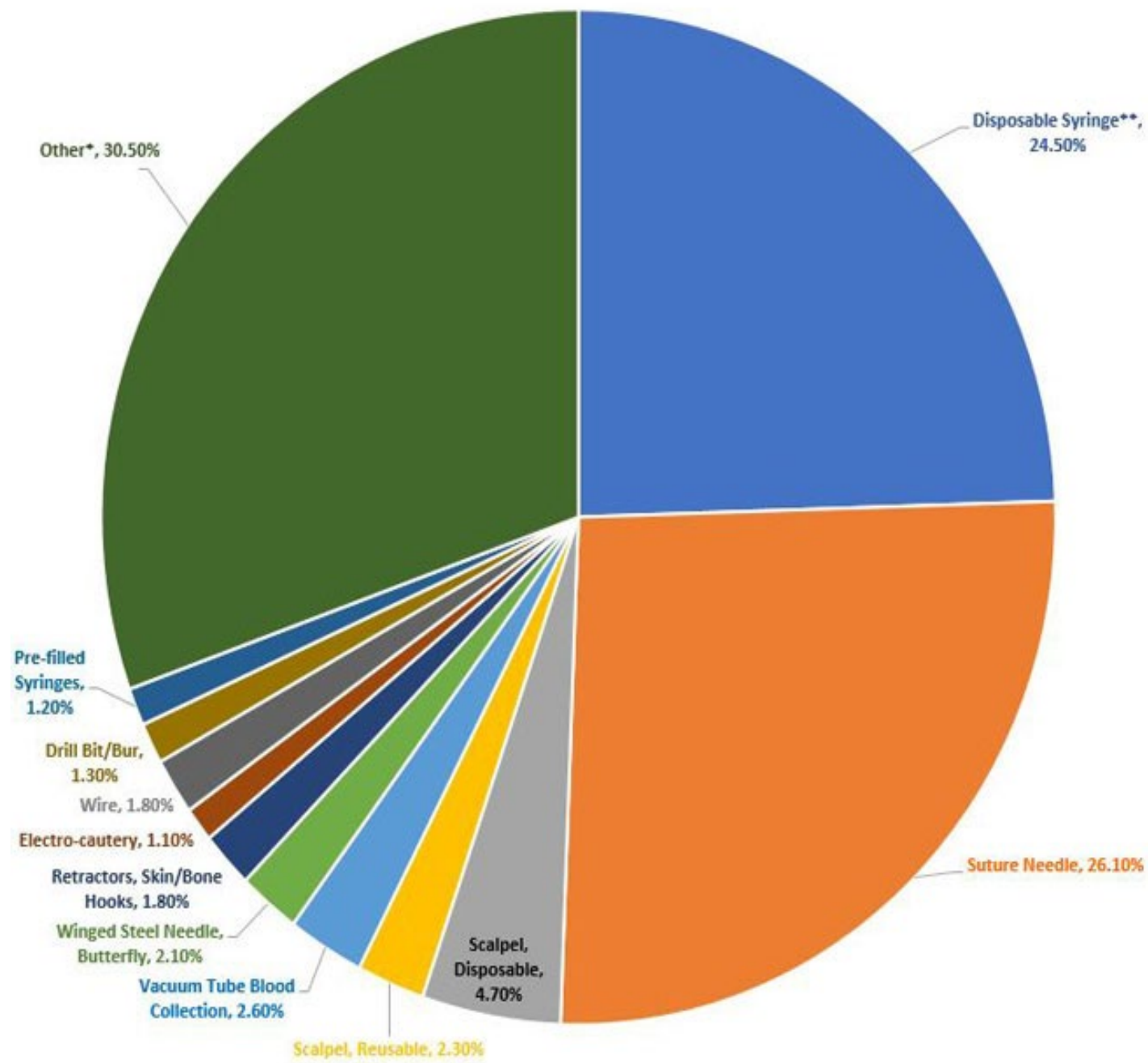
- Blades, needles, broken carpules or carpules contain blood
- Extracted teeth

Pharmaceuticals

- Unused/partially used medications
- Vials inhalers
- Carpules without blood

Handling Sharps

- Sharp instruments should be handled with caution
 - Wear PPE when handling sharp and hazardous materials
 - Keep the instruments covered until they are ready for use.
 - Never place the sharp portion of the instrument towards another person.
 - Place disposable sharps in a designated receptacle after use
 - Reusable sharps should be cleaned, placed in a sterilization bag, and sterilized with heat (placed in the autoclaved).



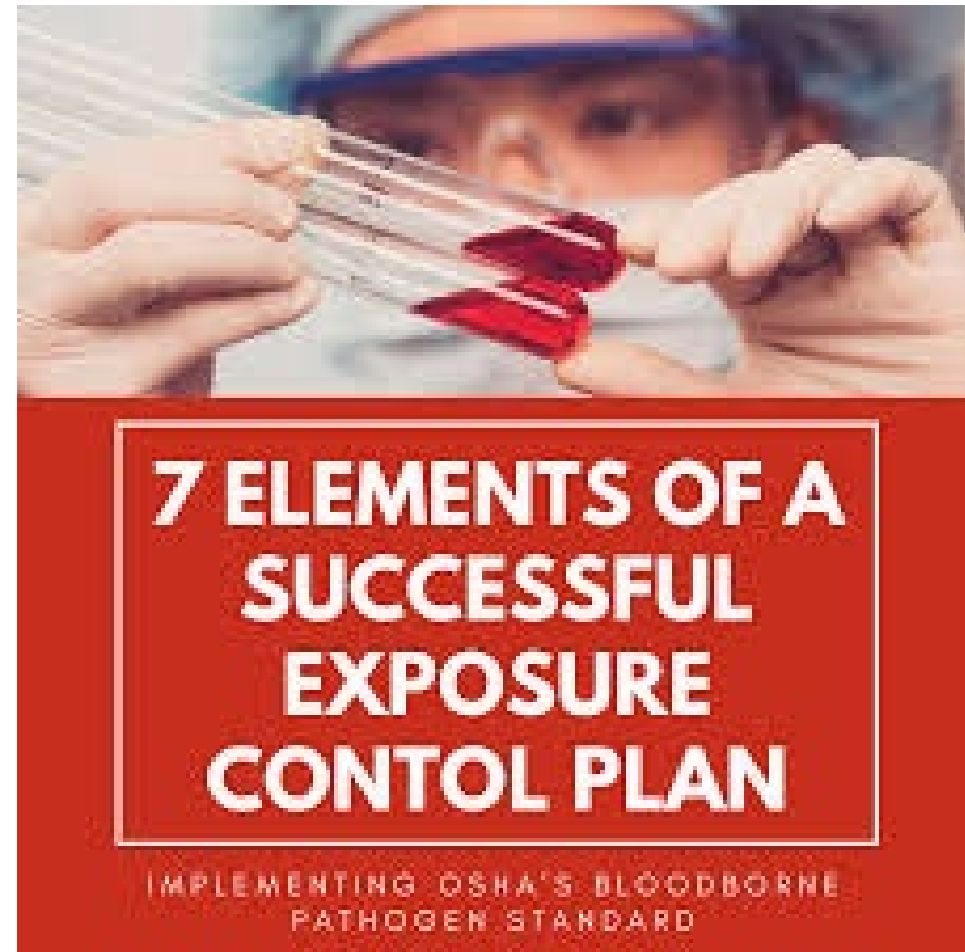
Needles usage

- Needles used for local anesthesia
 - Needles should NEVER be reused on multiple patients
 - Needles should remain covered until treatment is started
 - One handed scoop technique
 - Use of needle sheath



Exposure Plan

- Have a written plan
 - **Injury and Illness Prevention Plan**
 - **Include COVID-19 Prevention Plan**
 - Until Feb. 2026
- Provide training for all employees
- Follow all aspects of Standard Precautions
- Hepatitis B
- Influenza vaccine
- In the event of an exposure
 - File a written report
 - Ensure the proper testing is completed
 - Post-exposure prophylaxis (PEP)
 - Follow-up laboratory studies



Dental Offices and COVID-19

COVID-19 Prevention Nonemergency Regulations expired: What's next for dental practices?

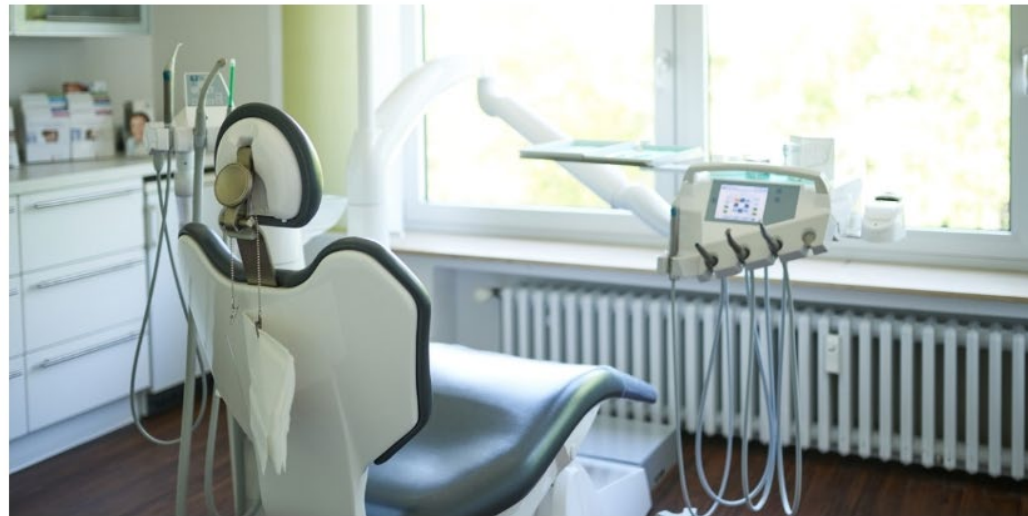
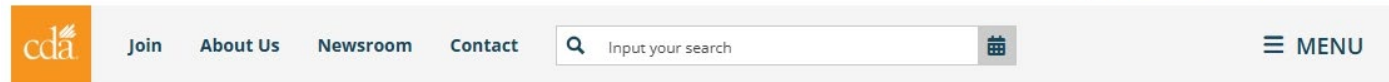
Employers' reporting and recordkeeping requirements remain in effect

Contact CDA

✉ contactcda@cda.org
☎ 800.232.7645

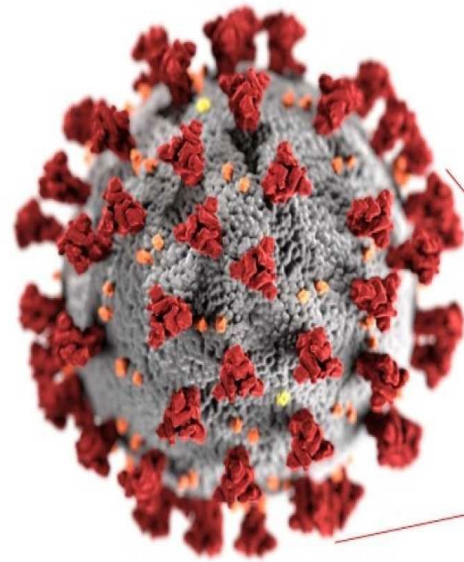
Related

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QUICK SUMMARY: With the nonemergency regulations' expiration, practice owners in California are no longer required to have a COVID-19 prevention plan or addendum to the injury and illness prevention plan but must continue complying with the reporting and recordkeeping requirements through Feb. 3, 2026.

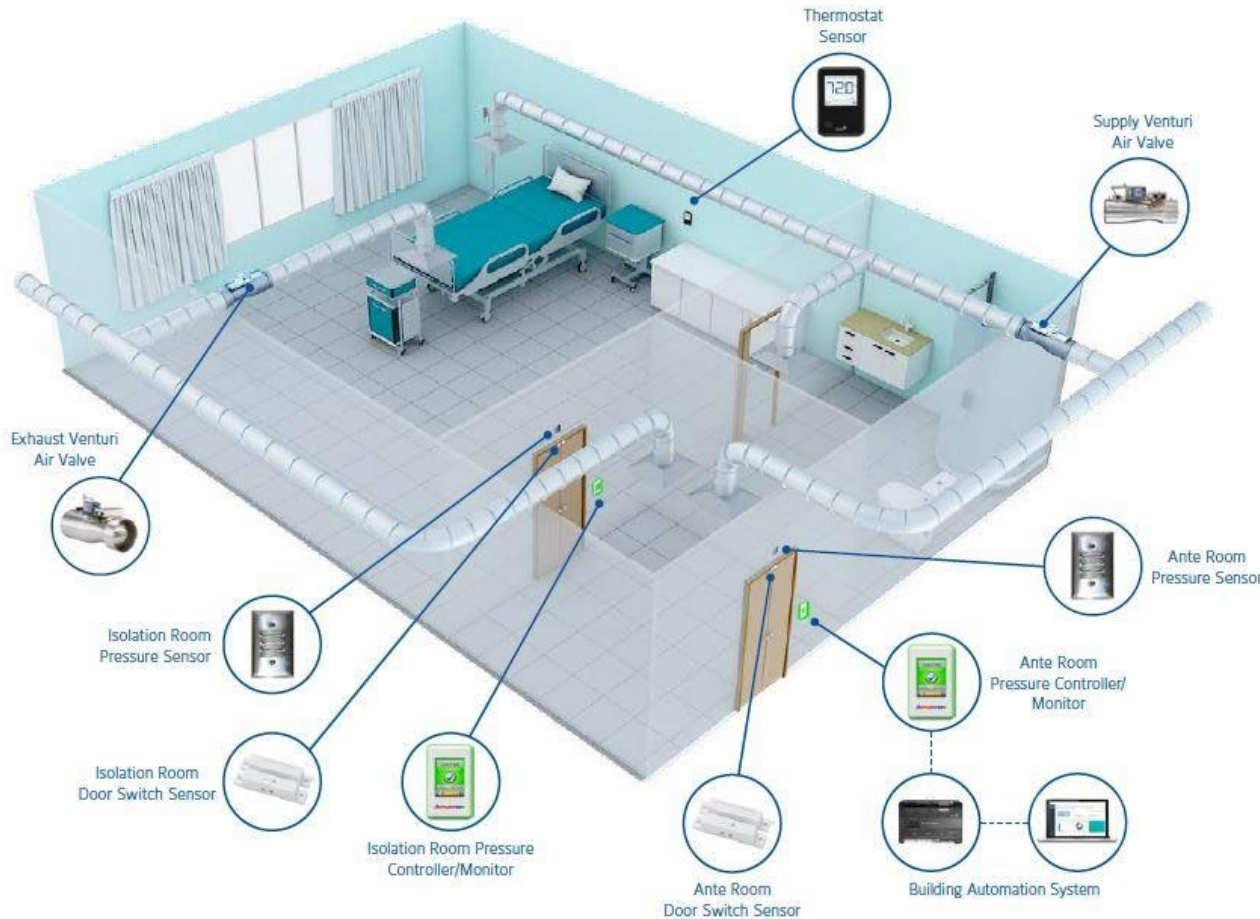
Coronavirus-containing aerosols



Powered Air-Purifying Respirator



Tuberculosis -All Room



- Patients with highly transmissible respiratory disease who require emergency treatment.
 - Tuberculosis
- Negative pressure isolation
- Health care workers in the room are required to wear N-95 mask.

What Happens After a Needle Stick?

- Remove gloves and wash exposed area with soap and water.
- Report and document the exposure.
- The clinician and the patient (source) must undergo lab testing for HIV, HBV and HCV provided by the employer.
 - **Post Exposure Testing Schedule**
 - Day of the incident
 - 4-6 weeks
 - 12 weeks
 - 6 months
- Keep the employee's medical record in a safe location. Post-exposure prophylaxis.

PrEP vs PEP

	
PrEP stands for Pre-Exposure Prophylaxis.	PEP stands for Post-Exposure Prophylaxis.
It means taking antiretroviral medicines (ART) <u>before</u> being potentially exposed to HIV to prevent becoming infected.	It means taking antiretroviral medicines (ART) <u>after</u> being potentially exposed to HIV to prevent becoming infected.

PEP: CDC 3-Step Risk Assessment

1. Type of exposure

- Intact skin vs un-intact skin
- Mucous membrane
- Percutaneous exposure vs superficial abrasion
- Hollow vs solid needle

2. Status of Patient

- HIV negative vs positive vs unknown

3. Decision on Treatment

- Determined by reviewing data and assessing risk

4. Begin regimen **ASAP, within 72 hours**

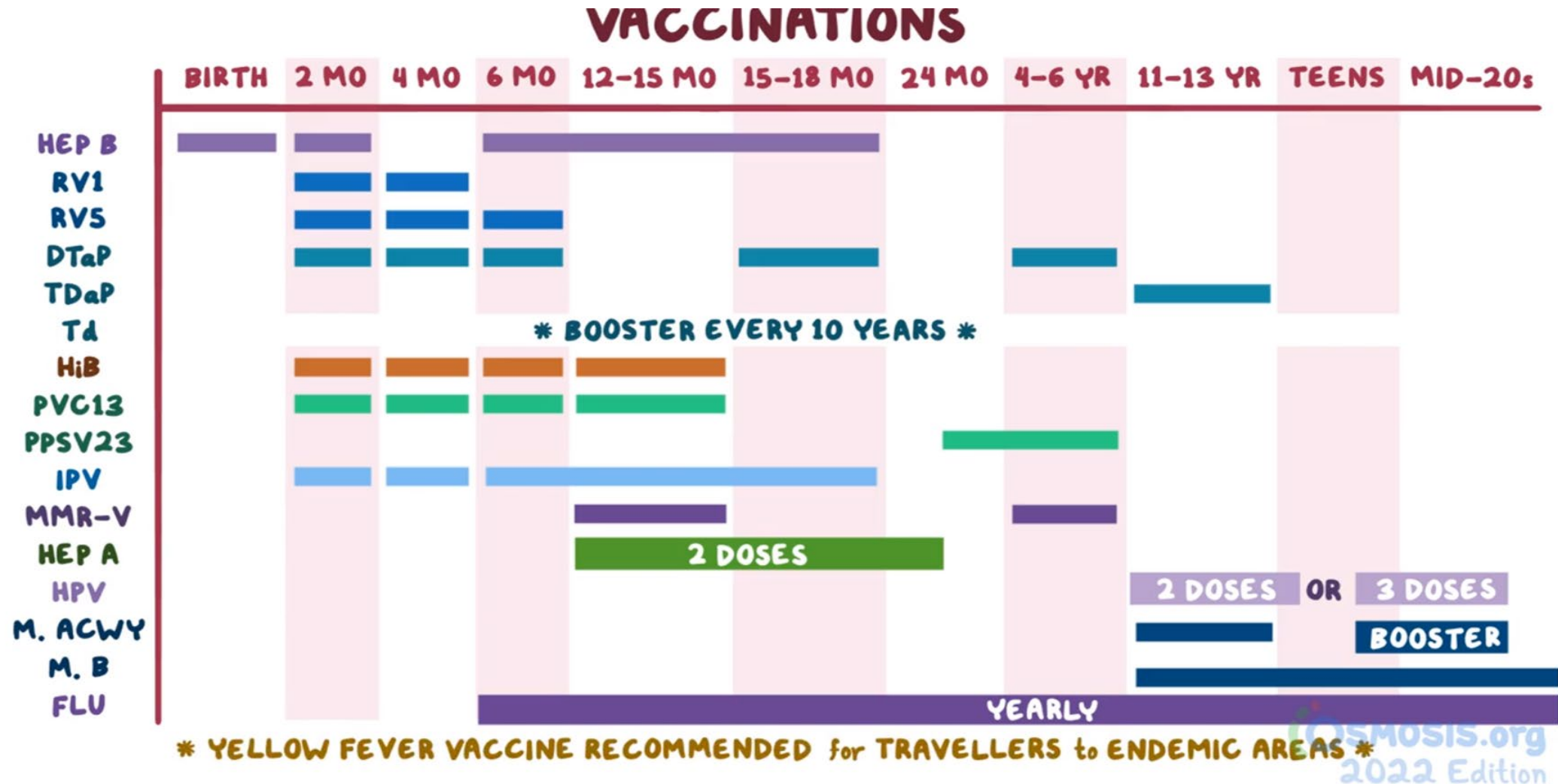
Vaccination Protocol

- Vaccination is required in some healthcare facilities and academic institutions.
- Regulations are influenced by the type of national, state, and local policies.
- Non-vaccinated employees are required to submit a waiver for certain vaccines.

Vaccine Administration

- Vaccines are an important type of primary prevention
 - Help prevent the outbreak of preventable infectious diseases
 - Disease prevention can help ease healthcare cost
 - Healthcare workers are given the opportunity to provide education and community awareness

Recommended Schedule for Vaccines

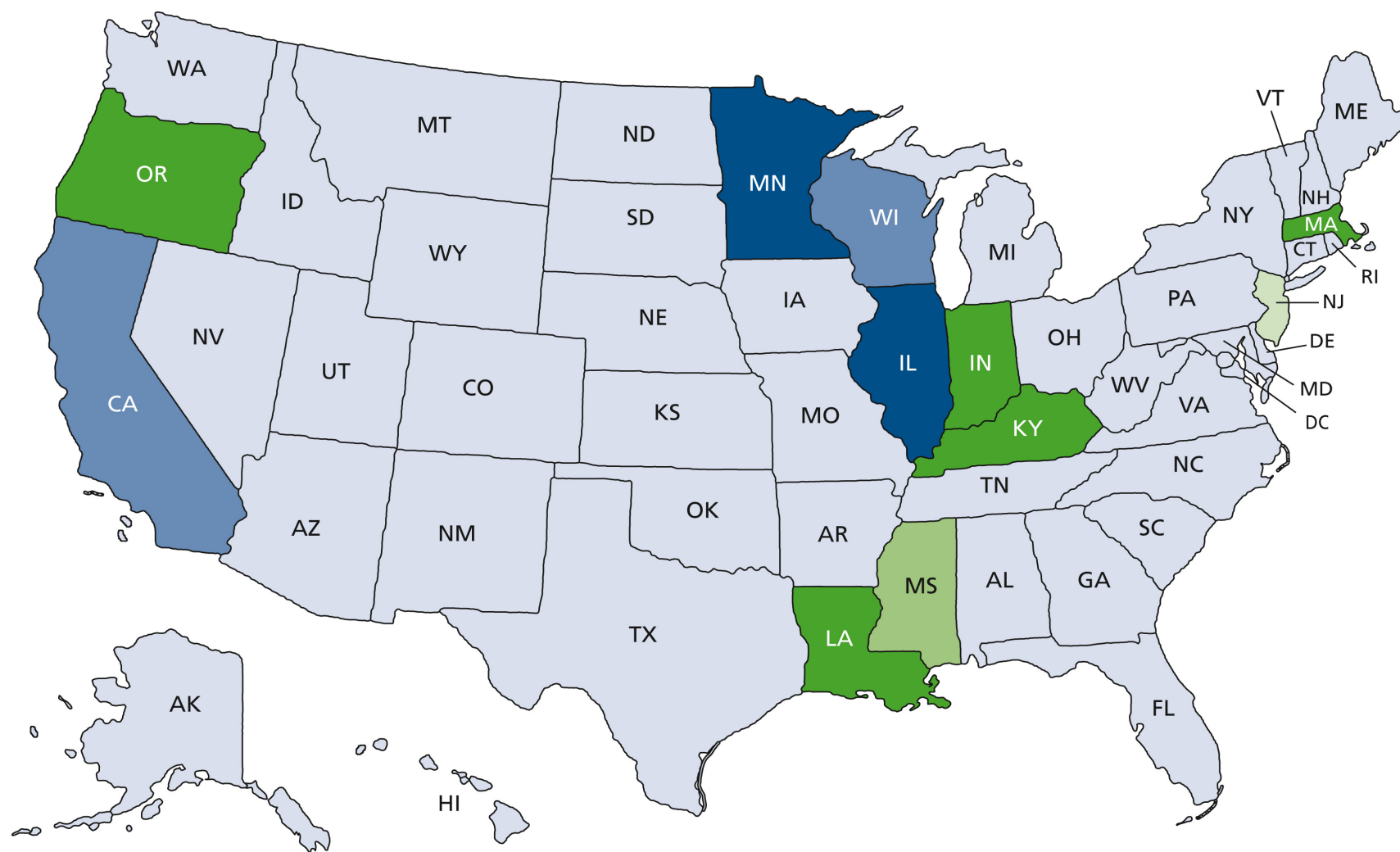


Vaccine Administration

- Vaccines are typically given by the healthcare providers in medicine.
 - Physicians, physician assistants, nurses, and nurse practitioners
- Pharmacist were given the right to provide vaccines starting in the 1990s through 2009.

Vaccine Administration in Dentistry

- Expanding the dental scope of practice
 - Allowing dentists to practice vaccines
 - Dentists in Illinois and Minnesota being allowed to give the influenza vaccines in starting in 2014 and 2016.
 - House Bill 2220 was signed allowing dentists in Oregon to provide routine vaccines in their dental practices.
 - January 2021- Dentists and dental hygienists providing COVID-19 vaccinations
 - 2024 – Variety of vaccines can be administered by dentists depending on the state.



State allows dentists to administer any vaccine

State allows dentists to administer COVID-19 and influenza vaccines

State allows dentists to administer influenza vaccine

State allows dentists to administer COVID-19, HPV, mpox, shingles, and smallpox vaccines

State allows dentists to administer HPV and influenza vaccines

Villa A, Saremi M, Klausner J et al. Oral health care practitioners as vaccine administrators. The Journal of the American Dental Association, 2023; 155, 556-558.e1

Advantages of Dentists Providing Vaccines

- Dentists tend to see patients more often when compared to physicians.
 - Vaccines can reach a larger group of people
- Dentists can provide education about the HPV-vaccine during the time of vaccination administration.
- Help develop interprofessional relationships between the medical and dental healthcare teams.

HPV VACCINE

* PROTECTS AGAINST HUMAN PAPILLOMA VIRUS

└ GENITAL WARTS, CERVICAL CANCER, & PENILE CANCER

Oropharyngeal Cancer

* 2 DOSES:

9 - 14 YEARS OLD → 6 MONTHS LATER



15 - 26 YEARS OLD → 1 - 2 MONTHS LATER → 6 MONTHS LATER

* NO SPECIFIC CONTRAINDICATIONS

└ NOT RECOMMENDED IN PREGNANT FEMALES

The vaccine can now be administered to patients up to the age of 45 years.

9 valent vaccine protects against 6, 11, 16, 18, 31, 33, 45, 52, and 58



Gardasil 9 Commercial (2024 & 2025)



Watch later



Share



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YouTube



Conventional vs HPV-related SCC

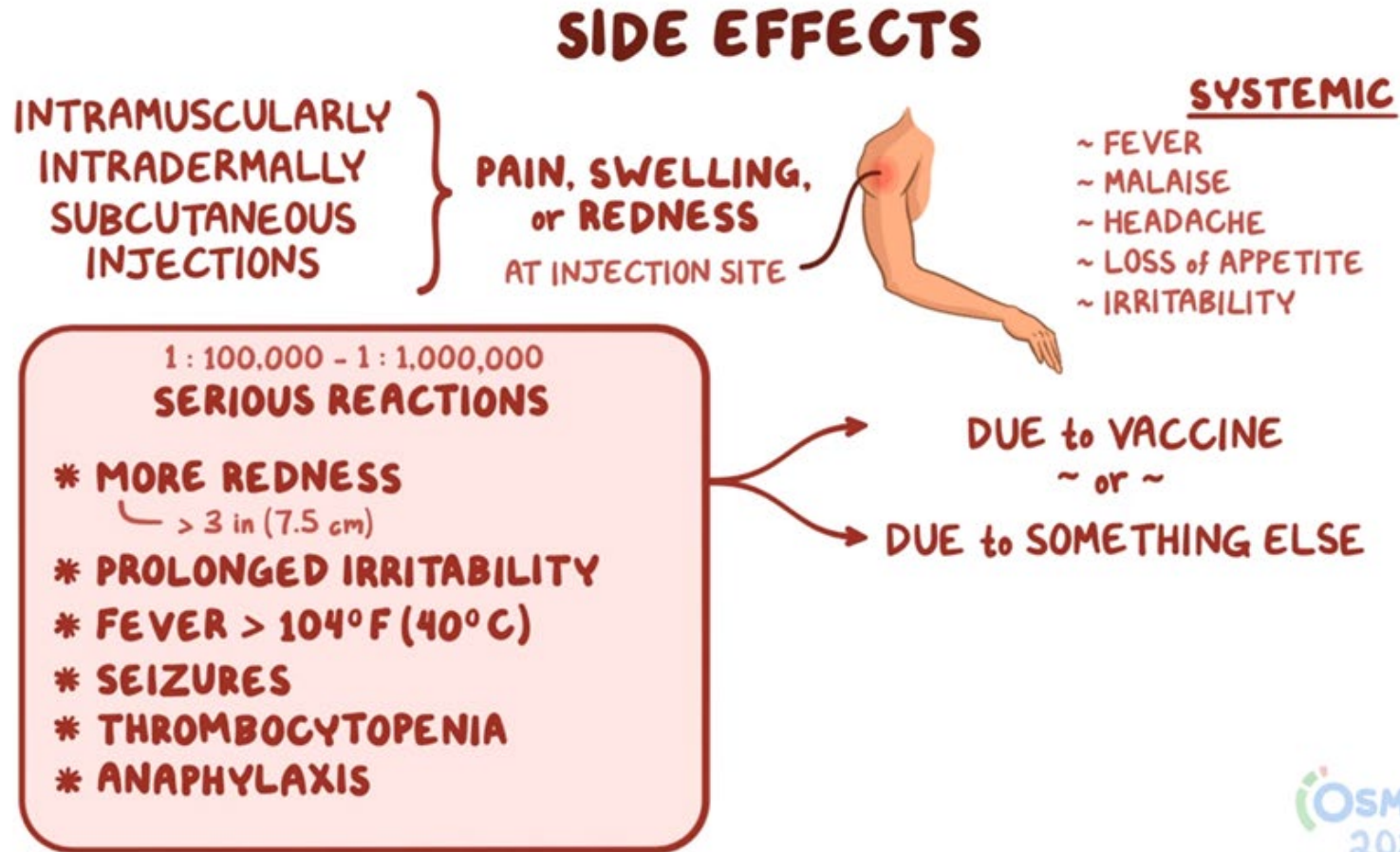


New England Journal of Medicine

Disadvantages of Dentist Giving Vaccines

- Dentists would need to have access to patients' health care records.
- The dentists would be responsible for addressing the side effects associated with vaccines.
- Insurance reimbursement.
- Time and resources dedicated to didactic and clinical training
- Dental providers level of comfort

Side Effects Associated with Vaccines



Big Picture

1. Know the governing bodies who set the guidelines for healthcare settings.
2. Know what bloodborne pathogens and risk of exposure to each in a dental setting.
3. Disease transmission of bloodborne and aerosolized diseases.
4. Prevention of disease transmission in healthcare settings.
 1. Be aware of policy changes in your area of practice and nationally.
5. The role dentists play in vaccination education and administration.

Resources

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16. [California Code of Regulations, Title 8, Section 5199 Appendix B.](#)
17. OSHA.org